

HAOLU MA

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GitHub: <https://github.com/Bend-Function>

CERTIFICATIONS

***AWS Certified Solutions Architect – Professional
OceanBase Version V2 Certification***

Project

Open-source Proxy Middleware — Owner

Built an open source proxy middleware that can connect to different proxy backends and proxy panels.

- Took full ownership of architecture and core code, adding support for multiple proxy protocols, automatic node self registration, user management, rate limiting, with one click deployment through Docker and GitHub Actions.
- The project has gained more than 740 stars and tens of thousands of downloads on GitHub, and is used in production with Ansible based automated deployment and health monitoring on more than 20 nodes.

WORK EXPERIENCE

Cloud Solution Architect | Remote

Sep 2023 – Aug 2024

Led the migration of an on premises and Azure hybrid environment to AWS, mapping system dependencies and designing production VPCs, subnets, routing and security groups aligned with AWS best practices.

- Designed and implemented load balanced and highly available core services across multiple Availability Zones, using health checks to achieve automatic failover and reduce downtime during incidents.
- Built a standard runtime based on custom business images so application services can scale up and down automatically across regions according to traffic and load, cutting manual operations during peak periods.
- Wrote migration and scaling runbooks and automation scripts, and trained the operations team to run and monitor the AWS environment confidently in day to day operations.

Hangzhou Baohang Industrial Investment Group Co., Ltd. | Hangzhou, China

Head of Software, R&D Department

Aug 2024 – Present

Led the software team to deliver a hotel wide IoT smart control system for Merchant Marco Hotel, covering 300+ rooms and 3,000+ devices, increasing staff productivity, cutting guest complaints by 50 percent and reducing energy costs by 10 percent after going live.

- Designed the overall architecture of the in house hotel IoT platform (React, Django, MySQL), from initial design to production, giving operations real time visibility, monitoring and control over all guest room and public area devices.

- Worked with the hardware team to create a custom IoT communication protocol and an MCP based orchestration layer that links in room voice assistants with large language models, so guests can use natural language while the system safely routes each request to the right device actions.
- Implemented queue and task scheduling services integrated with the hotel PMS to keep room status and service tickets in sync, and introduced a role based access control model that separates permissions for guests, front desk and operations teams and protects critical hotel systems.

K square tech | Hamilton, New Zealand

Software Engineer Intern

Feb 2024 – Jul 2024

Built and deployed an AWS based automation system for product and price management on an online store, covering data collection, AI processing and live updates into WooCommerce.

- Used Python, Scrapy and the OpenAI API to collect, clean and standardise supplier quotes into data ready for import, including automatic bulk uploads, and built a React admin dashboard to configure and monitor automation jobs and compare supplier price sheets.
- Set up a full CI/CD workflow with Gitea, TeamCity and YouTrack, then migrated it to AWS CodePipeline and ECS so test and production environments run with automated builds, deployments and releases.

Hangzhou Baohang Industrial Investment Group Co., Ltd. | Hangzhou, China

Data Scientist

Aug 2023 – Aug 2024

Led the digital transformation of the steel trading business, consolidating about 20 GB of ERP and finance data from three databases into a clean unified view for analysis.

- Built machine learning and deep learning models on internal and external steel trading and legal data to guide stock planning, price negotiation and risk control.
- Worked with sales managers and front line staff to map processes and turn business pain points into clear data and AI use cases, including an AI agent that searches public sources about new customers and returns concise risk and strategy suggestions.

EDUCATION

University of Auckland Master of Information Technology New Zealand	Feb 2025 - Mar 2026
Waikato Institute of Technology Bachelor of Applied Information Technology New Zealand	Jul 2023 - Jul 2024
Jinhua University of Vocational Technology Diploma of Computer Networking China	Sept 2019 - Jun 2022